

OAuth Authentication & Token Management

FinAccrual Login Tokens

These tokens are used to identify and authenticate the FinAccrual user.

- User logs in using:
 - Google
 - Microsoft Entra ID
- After login, we generate :
 - **Access Token** → used for normal API requests.
 - **Refresh Token** → used to get a new Access Token and refresh Token when the old one expires.

FinAccrual Access Token and Refresh Token

After successful login, two tokens are involved.

Token	Purpose	Lifetime
Access Token	Used to access FinAccrual APIs	Around 1 hour
Refresh Token	Used to generate a new Access Token and refresh Token	7 days

Access Token :

The Access Token is a **JWT**.

It contains information such as: User ID , Email , User role

It is sent with API requests through the Authorization header.

Refresh Token :

- The Refresh Token is an opaque/random token , It is used when the Access Token expires.
- The user does **not** need to log in again every time the Access Token expires. The Refresh Token can be used to generate a new Access Token.

What happens when the Access Token expires?

1. The user's Access Token expires or is close to expiring.
2. The frontend sends: POST /api/auth/refresh with the Refresh Token.

If the token is invalid or the user is inactive : 401 Unauthorized(Session Expired)

If the token is valid

1. Generates a new Access Token and Refresh Token, Updates the database.
2. Returns the new token pair to the frontend. So the user can continue working without logging in again.

When the Tokens clear in Database?

When the user logout, The system removes the stored authentication tokens from the database.

QuickBooks OAuth Authentication

- FinAccrual **does not store the user's QuickBooks password.**
- Instead, QuickBooks provides Access Token and Refresh Token
- These tokens allow FinAccrual to access the connected QuickBooks company.
- Each QuickBooks company has a unique:realmId
- The realmId is used to identify the connected company and its tokens.

QuickBooks tokens have different lifetimes.

Token	Purpose	Lifetime
Access Token	Access QuickBooks APIs	60 minutes
Refresh Token	Generate new QuickBooks tokens	Up to 100 days

Important point

Every time FinAccrual refreshes a QuickBooks Access Token, QuickBooks gives a **new Access Token and a new Refresh Token.**

The old Refresh Token becomes invalid.

Therefore, the system must always save **both new tokens.**

When the access Token Expires ?

QuickBooks Access Tokens expire after approximately **60 minutes**.

The system should not wait until the token completely expires.

Then backend sends the request to intuit with refresh token , then it verify and sends new tokens

Refresh Token Expiry Check

Before making a refresh request to QuickBooks, the system also checks:

Has the QuickBooks Refresh Token itself expired?

The Refresh Token is valid for approximately **100 days**.

If it has expired

1. Marks the company as Disconnected.
2. Clears the stored QuickBooks tokens.
3. Throws an authentication error.
4. Tells the user that QuickBooks needs to be connected again.

So the user must re-authorize the QuickBooks company.

So, simply:

FinAccrual token = Who is the user?

QuickBooks/Xero token = Which accounting company can the user access?